GNN Molecular Substructure Search: Accuracy vs. Speed Trade-offs Against Classical Fingerprint Screening

CS 6012: Deep Learning — Open Project (Submission 2: Final Report)

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GitHub: https://github.com/srijal20/GNN-Substructure-Search-project | **Colab Notebook:** Open in Colab

Abstract — *Exact molecular substructure search is an essential primitive in computational chemistry and virtual drug screening. Given a molecular query pattern Q, the goal is identifying all database compounds T containing Q as an induced subgraph ($Q \subseteq T$). Combinatorial algorithms (VF2, Ullmann) guarantee exactness but exhibit worst-case exponential time complexity $O(|V_T|^{|V_Q|})$. In this project, we implement and benchmark Graph Neural Network (GNN) architectures for approximate substructure filtering on ZINC250k: (1) a bond-aware Siamese Graph Isomorphism Network (GINEConv + MLP, BCE loss), and (2) a geometric order embedder based on NeuroMatch ($z_Q \leq z_T$ in $\blacksquare_{\geq 0}^{64}$). To satisfy real-world screening requirements where false negatives cannot be tolerated, we develop a **two-stage cascaded architecture**: the order-embedding GNN rapidly filters 50,000 molecules at a calibrated ≥98% recall threshold, followed by RDKit exact verification exclusively on the candidate shortlist. Our empirical findings show that while GNN order screening achieves an ultrafast 11.4× candidate retrieval speedup (2.79 ms vs. 31.84 ms brute force), its candidate pool (1,413.9 molecules vs. 703.3 for Pattern Fingerprints) creates an exact-verification bottleneck, causing overall cascaded latency (20.21 ms) to be outperformed by classical RDKit Pattern Fingerprints (5.61 ms, 100% recall). Furthermore, query size ablation reveals severe performance decay on large fragments (>83% recall for 4–5 atoms vs. 28%–44% recall for 10–12 atoms), exposing receptive-field and pooling bottlenecks in whole-graph sum-based representations.*

1. Introduction & Problem Statement

Molecular substructure retrieval evaluates whether a query chemical motif *Q* exists within a target molecular graph *T*. In formal graph theory, this is the NP-complete *subgraph isomorphism problem*. Cheminformatics libraries such as RDKit implement optimized tree-search algorithms (e.g., VF2, Ullmann) that perform sequential vertex-edge backtracking. While completely accurate, these routines become computationally intractable when repeatedly querying massive corporate or public libraries (such as ZINC250k or Enamine REAL with billions of compounds).

To accelerate search, cheminformatics relies on specialized bit-vector fingerprints, predominantly **RDKit Pattern Fingerprints**. By design, Pattern Fingerprints guarantee a monotonic bit-containment property: if $Q \subseteq T$, then (fp(T) & fp(Q)) == fp(Q). This bitwise screen achieves 100% recall with zero false negatives in ~1 ms, acting as a candidate filter before exact matching.

Research Objective: In this project, we explore whether deep Graph Neural Networks can learn chemical subgraph containment, whether offline target pre-embedding via geometric order embeddings can outperform bitwise fingerprint screens, and how the accuracy-speed Pareto frontier behaves when learned GNNs are placed in a realistic two-stage verification cascade.

2. Deep Learning Methodology & Architectures

2.1 Chemical Graph Featurization: We construct PyTorch Geometric graphs directly aligned with RDKit SMARTS matching rules:
• *Node Features:* 10-dimensional one-hot vectors for atomic numbers: C, N, O, F, P, S, Cl, Br, I, and an 'other' bucket.
• *Edge Features:* 4-dimensional one-hot vectors for bond orders: Single, Double, Triple, and Aromatic.
• *Topology:* Bidirectional edge connectivity preserving molecular valence.

2.2 Bond-Aware GIN Backbone (GINEConv): Standard message passing networks with mean/max aggregators fail to distinguish multiset structures. Following Xu et al. (ICLR 2019), sum aggregation provides maximal discriminative power equivalent to the 1-Weisfeiler-Lehman (1-WL) graph isomorphism test. We utilize 4 sequential GINEConv layers (Hu et al., 2020) with 2-layer internal MLPs and learnable ε parameters. To capture both local atom environments and broader ring motifs, we concatenate sum-pooled representations across all intermediate layers (0 to 4), yielding a 320-dimensional multi-scale vector before linear projection to 64 dimensions.

2.3 Model A (Siamese GIN + MLP Classifier): An unconstrained GINE encoder embeds query z_Q and target z_T. A 2-layer interaction classification head evaluates the concatenated features [z_Q, z_T, z_T - z_Q, z_Q * z_T] $\in \blacksquare^{256}$, trained with Binary Cross-Entropy (BCE) with logits loss.

2.4 Model B (NeuroMatch Geometric Order Embeddings): Inspired by Ying et al. (NeurIPS 2020), substructure containment naturally forms a partial order over graph space. By enforcing a non-negative activation (ReLU) on the final projection layer, embeddings are mapped to the non-negative orthant $\blacksquare_{\geq 0}^{64}$. Containment is enforced by element-wise dominance: $Q \subseteq T \Leftrightarrow z_Q \leq z_T$. The violation energy is formulated as:

$$E(Q, T) = ||max(0, z_Q - z_T)||_2^2$$

We optimize this using a max-margin ranking loss with margin $M = 1.0$:

$$L(Q, T) = y \cdot E(Q, T) + (1 - y) \cdot max(0, M - E(Q, T))$$

Crucially, target embeddings z_T across the entire database are precomputed once offline (0.67 s for 5,000 molecules). Online query search is reduced to a single vectorized PyTorch tensor subtraction on GPU.

3. Two-Stage Cascaded Retrieval Framework

Because standalone neural models produce false negatives (recall < 100%), they cannot directly replace exact matching in pharmaceutical screening. We structure our deployment into a **two-stage cascade**:
1. *Stage 1 (Neural Filter):* Model B evaluates $E(Q, T)$ against all pre-embedded database molecules and selects a candidate shortlist $S = \{T \in D \mid E(Q, T) \leq \tau_{casc}\}$. The threshold τ_{casc} is calibrated on a validation search to guarantee ≥98% recall.
2. *Stage 2 (Exact Verification):* RDKit's HasSubstructMatch verifies only compounds in shortlist *S*, guaranteeing 100% final precision.

4. Experimental Setup & Benchmark Results

Dataset & Split: We utilized ZINC250k (249,455 drug-like compounds). To ensure zero data leakage, we created four disjoint splits: 40,000 training molecules, 4,000 validation molecules, 50,000 search database molecules (completely unseen), and 2,000 held-out query sources. Queries consist of subgraphs of 4 to 12 atoms sampled via BFS, plus 20 realistic medicinal chemistry functional groups. Hard negatives were mined using pairs that pass fingerprint screening but fail exact topological matching.

Retrieval Method	Precision	Recall	F1-Score	Latency / Query	Speedup vs. Brute	Avg. Shortlist Size
RDKit Exact (Brute Force)	1.000	1.000	1.000	31.84 ms	1.0×	494.2
Pattern FP screen only	0.703	1.000	0.825	1.09 ms	29.2×	703.3
Pattern FP + RDKit (Industrial Baseline)	**1.000**	**1.000**	**1.000**	**5.61 ms**	**5.7×**	494.2
GNN A: GIN + MLP Classifier	0.461	0.638	0.535	6.66 ms	4.8×	683.2
GNN B: Order Embedding alone	0.270	0.771	0.399	**2.79 ms**	**11.4×**	1,413.9
Cascade: GNN B Shortlist + RDKit	**1.000**	0.982	0.991	20.21 ms	1.6×	485.5

5. Scientific Insights & Detailed Error Analysis

5.1 The Verification Bottleneck: While GNN B runs an order-embedding vector screen in just 2.79 ms (an 11.4× speedup), the cascade takes 20.21 ms. Why? To maintain ≥98% recall, the threshold must be set generously, which admits 1,413.9 candidates per query (compared to only 703.3 candidates for Pattern Fingerprints). Since RDKit verification takes ~0.012 ms per candidate, verifying 1,414 molecules takes ~17 ms of CPU time. *The critical bottleneck in neural retrieval is not screening speed, but candidate precision at high recall.*

5.2 Receptive Field Decay with Query Size:

We performed a fine-grained error analysis stratified by query atom count ($|V_Q| \in [4, 12]$ atoms):

Query Atoms ($\|V_Q\|$)	4	5	6	7	8	9	10	11	12
Order GNN Recall	**0.834**	**0.837**	0.681	0.699	0.786	0.678	**0.284**	0.435	**0.436**
Order GNN Precision	0.447	0.328	0.218	0.125	0.128	0.026	0.013	0.021	0.034
Mean True Hits in DB	1,573.7	599.8	787.0	330.4	280.3	51.1	10.0	32.7	18.2

Interpretation: For small fragments (4–5 atoms), recall is high (>83%). However, for larger 10–12 atom subgraphs, recall drops sharply to 28%–44%. Because our GINE network uses 4 message-passing layers, each node only gathers information within a 4-hop radius. When query graphs exceed 8–10 atoms (e.g., fused bicyclic ring systems), multi-hop topological cycles cannot be accurately resolved. Furthermore, global sum pooling compresses the graph into a fixed 64-dimensional vector, losing explicit subgraph node-mapping alignment.

6. Member Contributions & Project Division

- **Aayush Gupta (25m0140):** GPU acceleration & PyTorch CUDA runtime environment on RTX 3050; graph featurization pipeline (atom and bond one-hot encoders); GINEConv backbone, Model A (BCE classification), and Model B (NeuroMatch order embedding violation loss); ROC-AUC evaluation metrics; high-recall threshold optimization algorithm; final report authoring and video script preparation.
- **Srijal Gupta (25m0387):** ZINC250k data loading and disjoint dataset partitioning; breadth-first connected query subgraph generation (4 to 12 atoms); functional group query curation (20 chemical classes); hard negative pair mining against fingerprint screens; RDKit Pattern Fingerprint and exact baseline evaluation harness; query-size error ablation; schedule tracking.
- **Joint Deliverables:** Cascaded pipeline formulation, Pareto frontier trade-off analysis, Colab notebook verification, and final submission.

7. Conclusion

This work provides an honest, empirical evaluation of GNNs for molecular substructure retrieval. While NeuroMatch geometric order embeddings are conceptually elegant and allow precomputed database screening in 2.79 ms (an 11.4× retrieval speedup), the classical RDKit Pattern Fingerprint remains a vastly superior filter in practice (5.61 ms at 100% precision and 100% recall). For deep learning to supersede classical bitwise filters, future models must incorporate higher-order subgraph-level message passing and achieve near-zero false-positive rates at the 98% recall boundary.

Key References: [1] K. Xu et al., *How Powerful are Graph Neural Networks?*, ICLR 2019. [2] R. Ying et al., *Neural Subgraph Matching*, NeurIPS 2020. [3] D. Rose et al., *PharmacoMatch*, ICLR 2025. [4] I. Vendrov et al., *Order-Embeddings of Images and Language*, ICLR 2016. [5] G. Landrum, *RDKit: Open-source cheminformatics*, 2026.